Salt bridge

The electrolytes present in two half cells are connected using a salt bridge. We have learnt that the anodic oxidation of zinc electrodes results in the increase in concentration of Zn^{2+} in solution. i.e., the solution contains more number of Zn^{2+} ions as compared to SO_4^{2-} and hence the solution in the anodic compartment would become positively charged. Similarly, the solution in the cathodic compartment would become negatively charged as the Cu^{2+} ions are reduced to copper i.e., the cathodic solution contain more number of SO_4^{2-} ions compared to Cu^{2+}.

To maintain the electrical neutrality in both the compartments, the non reactive anions Cl^- (from KCl taken in the salt bridge) move from the salt bridge and enter into the anodic compartment, at the same time some of the K^+ ions move from the salt bridge into the cathodic compartment.

Completion of circuit

Electrons flow from the negatively charged zinc anode into the positively charged copper cathode through the external wire, at the same time, anions move towards anode and cations are move towards the cathode compartment. This completes the circuit.

Consumption of Electrodes

As the Daniel cell operates, the mass of zinc electrode gradually decreases while the mass of the copper electrode increases and hence the cell will function until the entire metallic zinc electrode is converted in to Zn^{2+} or the entire Cu^{2+} ions are converted in to metallic copper.

Unlike Daniel cell, in certain cases, the reactants (or) products cannot serve as electrodes and in such cases inert electrode such as graphite (or) platinum is used which conducts current in the external circuit.

9.3.2 Galvanic cell notation

The galvanic cell is represented by a cell diagram, for example, Daniel cell is represented as

$$Zn\ (s)\,|\,Zn^{2+}(aq)\,\|\,Cu^{2+}(aq)\,|\,Cu\ (s)$$

In the above notation, a single vertical bar (|) represents a phase boundary and the double vertical bar (||) represents the salt bridge.

The anode half cell is written on the left side of the salt bridge and the cathode half cell on the right side.

The anode and cathode are written on the extreme left and extreme right, respectively.

The emf of the cell is written on the right side after cell diagram.

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